MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (GENERIC)

END OF SEMESTER EXAMINATION

ICT 4702 : MODELING AND SIMULATION

TIME ALLOWED: 3 HOURS **DATE : 6th APRIL , 2017**

INSTRUCTIONS

i. This paper contains **ONE**page. Please check.
ii. There are **TEN QUESTIONS**in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

1. Define the following terms:

   a) Performance. **[2 Marks]**
   b) A Model. **[2 Marks]**
   c) Cellular Automata. **[2 Marks]**
   d) Random-number Generator Seed. **[2 Marks]**
   e) Queuing Network. **[2 Marks]**

2. Describe Von-Neumann´s mid -square method and the congruential method for generating random numbers. For each method, give an example where the generation is considered to be poor.

   [10 Marks]

3. Describe the Kendall notation for queues. **[10 Marks]**

4. Consider a wireless access gateway:
   - Measurements have shown that packets arrive at a mean rate of 125 packets per second, and are buffered.
   - The gateway takes 2 milliseconds on average to transmit a packet.
   - The buffer currently has 13 places, including the place occupied by the packet being transmitted and packets that arrive when the buffer is full are lost.

   Making exponential assumptions about the arrival rate and the service rate, model the gateway using Kendall's Notation and calculate the traffic intensity, **ρ**.

   [10 Marks]

5. Explain how Simulation is applied in aircafts and hospitals **[10 Marks]**

6. The distinction between model verification and model validation is not always clear in practice. If you were told that 'this Discrete-Event Model has been verified but it is not known if the model is valid', how would you interpret that statement?

   [5 Marks]

7. By tackling all the important aspects, describe the single server model. **[10 Marks]**

8. Given knowledge of arrival times $a_1, a_2, a_3, \ldots$ and associated service times $s_1, s_2, s_3, \ldots$ and the queue discipline. How can delay times $d_1, d_2, d_3, \ldots$ be computed?

   [10 Marks]

9. If the queue discipline is FIFO, the node capacity is infinite and the server is idle both initially (at t=0) and immediately after the departure of nth job (at t=c_n), prove that

$$\int_0^{c_n} l(t)dt = \sum_{i=1}^{n} w_i$$

[15 Marks]

10. Explain Channel Assignment in cellular call setup in relation to Cellular Automata.
[10 Marks]

END OF QUESTION PAPER